

15. Howe Bay

Shoreline length: 23 km



Shore Types

Howe Bay SMU includes an open coastline facing East and an estuary protected by a barrier sandspit, with a majority of low plains, cliffs, and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
40	5	0	29	21	0	0	5



Barrier sandspit at Howe Bay entrance



Head of Howe Bay estuary

Governance, Infrastructure, and Economy

Approximately 43% of this SMU is unincorporated; the remaining 57% lies within the Rural Municipality of Annandale-Little Pond-Howe Bay.

Coastal infrastructure includes Grams Beach Road towards the sandspit, Chads Road at the beach end on the north side, and Route 310. There are no significant coastal economic drivers identified. Local and legacy land use constraints include few small lots on private roads along the coast.

Natural Assets and Heritage

There are no Provincial or Federal Parks, nor Wildlife Management Areas, in this SMU. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow, notably along the outer cliffs and the sand spit.

Coastal Flood and Erosion Risk

While low plains and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. The erosion hazard is highest along the exposed east shoreline. Erosion hazard inside the estuary is limited by the sheltered exposure and barrier sandspit.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline]

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	0	1 [<1/km]	2 [<1/km]	3 [<1/km]	0	2 [<1/km]	10 [<1/km]
Roads	168 m [7 m/km]	287 m [13 m/km]	328 m [14 m/km]	479 m [21 m/km]	30 m [1 m/km]	57 m [3 m/km]	113 m [5 m/km]

Planning and Policy Framework

Both the municipal and unincorporated areas in this SMU fall under provincial planning authority. Provincial rules apply throughout this SMU. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions.

There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Sediment transport

Sediment transport potential is highest along the exposed cliffs/ bluffs, which feed the barrier spit. It is lower inside the estuary.

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
SMU 15.00 Howe Bay	NAI	MR	MR
SMU 15.01 Howe Bay Head of Estuary	NAI	NAI	MR
Rationale	It is critical to maintain sediment transport towards the sandspit protecting the bay. Given the low risk levels the proposed strategy is initially No Active Intervention. Managed Realignment will then reduce risk to the limited number of vulnerable assets and the need for future intervention.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.